POLICY BRIEF

Strengthening community-managed irrigation systems in Gilgit-Baltistan

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Policy proposition

Strengthen community-managed irrigation systems in Gilgit-Baltistan

Gilgit-Baltistan should reform irrigation governance by formalizing existing customary community-based water institutions as Local Water Management Utilities (LWMUs), combining traditional water-management arrangements with modern technical, financial and administrative systems.

1. Executive Summary

Irrigation is the foundation of agriculture and rural settlement in Gilgit-Baltistan, where cultivation depends almost entirely on snowmelt- and glacier-fed systems. More than 90 percent of traditional irrigation infrastructure is managed through customary community arrangements. These systems remain resilient, but aging infrastructure, climate hazards, labor migration, weak routine financing, fragmented public mandates, and limited technical support are placing them under increasing strain.

The study concludes that sustainable reform should strengthen rather than replace community management. It recommends formalizing customary arrangements as Local Water Management Utilities: hybrid organizations that preserve legitimate traditional rules while adopting modern bylaws, professional O&M, full-time or shared technical capacity, transparent financial management, user service charges, and public subsidies. The Utilities should operate within the Local Government system, supported by the Irrigation Department, GBDMA, and other provincial institutions.

2. Why this is important for Gilgit-Baltistan?

Gilgit-Baltistan's unique geography at the junction of the Himalaya, Karakoram, and Hindukush mountain Gilgit-Baltistan’s high-mountain environment receives little rainfall, and rain-fed agriculture is virtually absent. Communities have therefore constructed and maintained extensive gravity-fed channels across steep and unstable terrain. Customary institutions regulate water turns, mobilize collective labor, resolve disputes, and repair local systems. Irrigation is consequently not only an infrastructure service; it is the basis of food production, livelihoods, settlement, and local cooperation.

This model is now under pressure. Outmigration and livelihood diversification have reduced voluntary labor, while floods, landslides, GLOFs, sedimentation, and changing seasonal flows are increasing the frequency and technical complexity of repairs. Government engagement remains concentrated on

capital projects and post-disaster rehabilitation. Without institutional reform, routine maintenance will weaken, service interruptions will grow, and public spending will remain reactive rather than preventive.

3. Key evidence

Finding	Key evidence
Community management	More than 90% of traditional irrigation infrastructure is reported as community-owned and managed through customary rules for allocation, maintenance, labor, and dispute resolution.
Episodic public support	Government engagement is concentrated in construction, rehabilitation, and disaster response, with limited routine support across the asset lifecycle.
O&M under strain	Migration, aging populations, rising cash needs, and complex repairs are weakening reliance on voluntary collective labor alone.
Increasing climate exposure	Aging channels and intakes face floods, landslides, GLOFs, sedimentation, and seasonal-flow variability that increasingly exceed local capacity.
Limited technology adoption	Modern and digital technologies remain constrained by collective incentives, subsistence production, maintenance capacity, and weak institutional support.
Incomplete evidence base	Systematic data on assets, performance, water losses, maintenance costs, financing, and technology outcomes remain limited.

4. Policy gaps

(a) **Legal and institutional gap:** customary bodies manage most systems but lack a consistent legal form that protects local authority while enabling professional staff, contracts, accounts, service charges, subsidies, and public oversight.

(b) **Local-government gap:** irrigation service management is not fully embedded within the Local Government system, and routine O&M lacks predictable local planning and budget provisions.

(c) **O&M financing and capacity gap:** voluntary labor remains important, but Utilities do not yet have dependable technical personnel, preventive-maintenance systems, recurrent revenues, or transparent subsidy arrangements.

(d) **Climate-risk gap:** vulnerability assessment, preparedness, and restoration arrangements are not systematically integrated with GBDMA and district disaster-management systems.

(e) **Technical stewardship gap:** the Irrigation Department requires stronger mountain-irrigation design, technology appraisal, climate-resilient engineering, and continuing technical-backstopping capacity.

(f) **Information gap:** no common asset, service, finance, risk, and performance information system connects communities, Local Governments, and the Irrigation Department.

5. Priority recommendations

5.1 Establish Local Water Management Utilities: Formalize customary irrigation bodies as legally recognized hybrid public-community Utilities. Document traditional rights, water turns, labor obligations, sanctions, and dispute-resolution practices and incorporate them into modern bylaws covering governance, service standards, staffing, finance, audit, inclusion, and grievance handling. Review the Water User's Associations Act, 2024 and Local Government legislation to determine whether amendments, subsidiary rules, or new legislation are required.

5.2 Integrate the Utilities into Local Government: Place recognition, regulation, oversight, and recurrent budget support within the Local Government system while preserving community ownership and operational authority. Define the mandates of Local Governments, LG&RD, the Irrigation Department, Finance, P&D, Agriculture, district administration, and GBDMA. Establish rules-based local budget grants for O&M, resilience, and major repairs.

5.3 Professionalize routine O&M: Make regular and preventive O&M the Utilities' primary mandate. Require annual operating and maintenance plans, asset and repair records, inspections, water-distribution management, and rapid fault response. Each Utility should employ, share, or contract qualified technical and management staff. Recurrent services should be financed through locally determined user service charges, recorded community contributions where appropriate, and predictable public subsidies reflecting affordability, remoteness, climate exposure, and technical complexity.

5.4. Institutionalize climate-risk management: Require each Utility to prepare and periodically update a vulnerability and risk assessment with GBDMA, district authorities, Local Government, and the Irrigation Department. Translate assessments into prioritized resilience works, preventive actions, contingency plans, emergency responsibilities, and restoration protocols, backed by dedicated public financing for works beyond local capacity.

5.5. Strengthen Irrigation Department technical capacity: Position the Irrigation Department as the province-wide technical steward. Build specialist capacity in mountain hydrology, geotechnical risk, intake and channel design, storage, gravity-fed piping, solar lift systems, appropriate micro-irrigation, and climate-resilient engineering. Provide Utilities with accessible diagnostics, designs, procurement specifications, construction supervision, technology appraisal, and support for complex repairs.

5.6. Establish an irrigation information and learning system: Create a georeferenced registry and performance system jointly maintained by the Irrigation Department, Local Governments, and Utilities. Record assets, users, condition, hazards, service reliability, failures, maintenance costs, user charges, subsidies, water losses where measurable, technology performance, and rehabilitation history. Use the evidence to prioritize investment, benchmark services, target technical support, and refine policy.

6. Implementation roadmap

Implementation should be phased and learning-oriented rather than imposed province-wide at the outset.

Phase	Principal actions
Phase-1: Foundation	Document customary systems; review legislation; define the Utility legal form, model bylaws, Local Government oversight, departmental roles, staffing norms, service-charge and subsidy rules, safeguards, and common data standards.
Phase-2. Pilot establishment	Select approximately ten diverse district sites; legally constitute Utilities; recruit or contract technical capacity; prepare O&M, financial, asset, inclusion, and vulnerability plans; and establish baseline data.
Phase-3: Operational testing	Implement professional O&M, charge and subsidy systems, local-government oversight, Irrigation Department technical support, GBDMA coordination, and routine performance reporting.
Phase-4: Evaluation and adaptation	Assess service reliability, maintenance performance, finances, affordability, inclusion, risk preparedness, institutional mandates, and technical support. Revise legislation, bylaws, staffing, and financing formulas.
Phase-5: Gradual scaling	Extend proven arrangements through the Local Government system while allowing local bylaws and operating practices to reflect legitimate customary variations.

7. Expected outcomes

(a) Reliable irrigation services: more systematic preventive maintenance, faster repairs, and fewer prolonged seasonal interruptions.

(b) Stronger local institutions: customary legitimacy preserved within accountable, legally recognized, professionally managed Utilities.

(c) Sustainable O&M financing: a transparent blend of user service charges, community contributions, and predictable public subsidies.

(d) Climate-resilient infrastructure: risk-informed investment, stronger preparedness, and quicker restoration following floods, landslides, and GLOFs.

(e) Better public stewardship: clearer mandates for Local Government, the Irrigation Department, GBDMA, and other agencies, with less reliance on ad hoc projects.

(f) More appropriate technology: designs and technologies selected according to system needs, lifecycle costs, downstream effects, and local maintenance capacity.

(g) Evidence-based investment: reliable asset, service, finance, and risk data guiding prioritization, budgeting, technical support, and continuous policy learning.

(h) Improved long-term development outcomes: more secure agricultural production, livelihoods, food security, social cooperation, and sustainable management of mountain water resources.

Source

This policy brief is based on the Irrigation Study Technical Report (Draft, May 2026), prepared under the EU Policy Support Technical Assistance to the Government of Gilgit-Baltistan. Statistics, recommendations and the implementation roadmap are drawn from the source document, subject to any future updates as improved regional datasets become available.

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